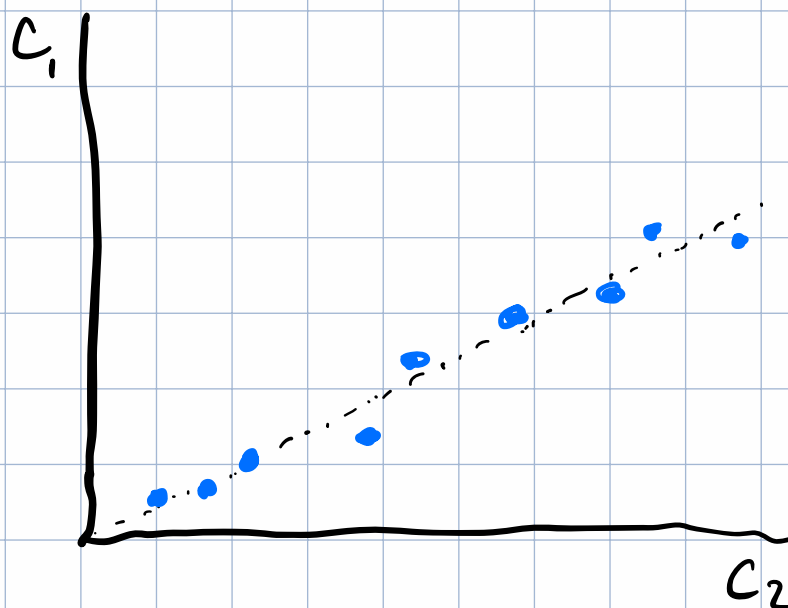


1) If the covariance between two variables is strongly positive, what does that mean geometrically?

It means that those two variables trend together. It also says that the slope of the line is positive. The data points should form a well-defined line.

Something like this:



2) How come the diagonals of the covariance matrix is always symmetric?

Algebraically, the $\text{Cov}(A) = \frac{1}{n} A^T A$, which is symmetric.

3) Representationally, variance and covariance are describing how two variables vary together. Variance is just the special case of how a variable varies with itself.

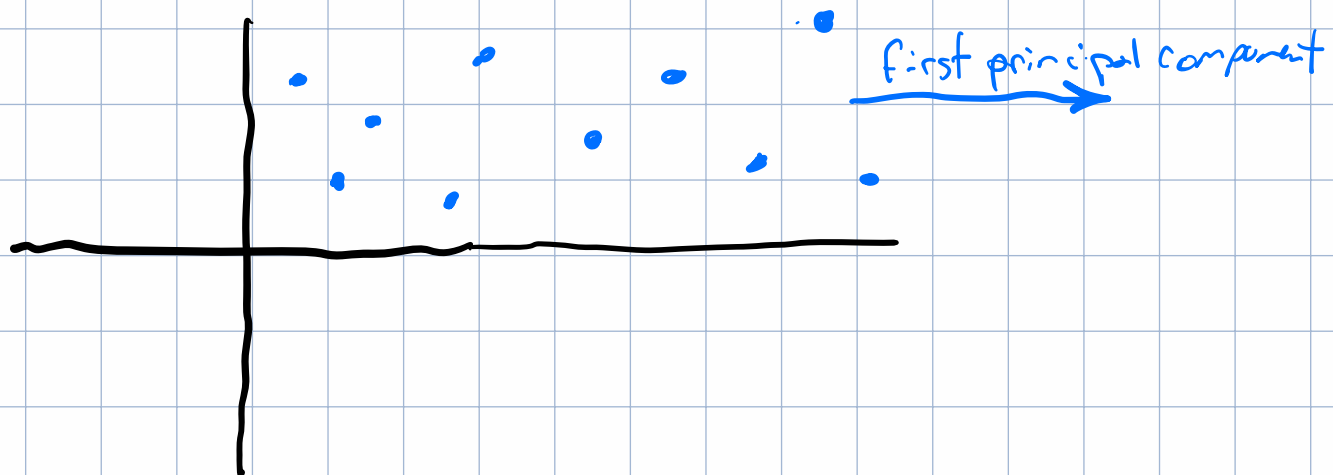
↑
Answers exercise 3

For the case $n=4$

$$\begin{aligned} x_0 &= \text{red dot} \\ x_1 &= \text{yellow dot} \\ x_2 &= \text{green dot} \\ x_3 &= \text{blue dot} \end{aligned} \quad \Sigma^{\text{Cov}} = \begin{bmatrix} \text{red, red} & \text{yellow, red} & \text{green, red} & \text{blue, red} \\ \text{red, yellow} & \text{yellow, yellow} & \text{green, yellow} & \text{blue, yellow} \\ \text{red, green} & \text{yellow, green} & \text{green, green} & \text{blue, green} \\ \text{red, blue} & \text{yellow, blue} & \text{green, blue} & \text{blue, blue} \end{bmatrix}$$

$\text{Cov}(x, y)$ commutes, because $A^T A$ commutes.

4) Suppose $\Sigma = \begin{bmatrix} 16 & 0 \\ 0 & 1 \end{bmatrix}$, sketch the approximate shape of data cloud. Which direction is the first principal component.



The variance is high for the first variable, and the variance is much lower for the second variable. The two variables are independent because they have zero covariance.

5) Why does principal component analysis compute the eigenvectors of the covariance matrix, rather

than the original data matrix?

Because we are finding the directions of the highest variance after it has been reoriented into the principal coordinate system. The covariance matrix tells us how the data relates to itself, including its variation. The eigenvectors of the covariance matrix are oriented in the direction of the highest variance.

Eigenvectors require a square matrix, and the data matrix is unlikely to be square. If we by some chance had equal-dimensional data, the eigenvectors would not have much meaning.